**REBECCA GEORGE**
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## EDUCATION:

**William and Mary -** B.S. in Computer Science, May 2026

- *Secondary major:* Computational and Applied Mathematics and Statistics (CAMS), Mathematical Biology track
- *Awards*: Dean's List, Phil Kearns Computer Systems Award

**Certificates**:

- Ubuntu Linux Professional Certificate by Canonical, Docker Foundations Professional Certificate

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## EXPERIENCE:

**William and Mary** — Williamsburg, VA
Undergraduate Research Assistant (Computer Science) — June 2025 - May 2026

- Wrote C++ scripts to benchmark solve times of GPU-accelerated vs. CPU-only versions of sparse linear solvers
- Ran I/O benchmarks on Argonne National Lab's Aurora supercomputer DAOS storage system, evaluating bandwidth, latency, and IOPS and using Bash to automate jobs
- Visualized results with Matplotlib/Seaborn/Pandas in Python; Presented research at CAPWIC '26

**Thomas Jefferson National Accelerator Facility** — Newport News, VA
Student Technical Intern — September 2024 - May 2026

- Installed and troubleshot Mac, Windows, and RedHat Linux operating systems, file backups, applications, and remote access
- Traced logs, ran diagnostic tests, and identified root causes of diverse issues (system, drivers, network, account, access, etc)
- Completed many tasks at once, communicating effectively with users and team members in fast-paced environment
- Managed accounts and authentication methods
- Wrote documentation and reference books for the helpdesk
- Maintained PowerShell scripts automating procedures
- Used Excel to track and report statistics on group operations
- Supported data center operations including hardware deployment and tape library management
- Wrote Puppetry supporting scientific computing infrastructure

**William and Mary** — Williamsburg, VA
Undergraduate Research Assistant (Mathematical Plant Ecology) — August 2023 - May 2024

- Integrated real genetic identities into a data-driven milkweed population model to characterize clonal population structure, enabling demographic inference across genets
- Performed model selection for generalized linear mixed-effects models (GLMMs) in R to analyze reproductive trait variation (flowering, seed pod production), evaluating fit via AIC and residual diagnostics
- Created interactive plotly dashboard to explore clonal reproduction characteristics of milkweed using Python
- Prepared milkweed leaf samples for genetic analysis

**FIRST Robotics Competition Team 2881** — Austin, TX
Robotics Instructor — July 2018 - August 2024

- Led annual STEM/robotics camps for elementary age Girl Scouts, coordinating volunteer teams and operations
- Developed lesson plans and taught programming and mechanical design with LEGO robots

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## SKILLS:

- Soft Skills: Problem solving, teamwork, time management, communication, troubleshooting, documentation
- Programming: R, Python, C/C++, Bash, Powershell, MATLAB, HTML/JS/CSS, Java
- Misc Software: Linux, Git, Markdown, LaTeX, Google Workspace, Microsoft Office, Jupyter Notebook
- Design/Mechanical: Solidworks, Bricklink Studio, Procreate
- Languages: English (fluent), German (conversational), Mandarin Chinese (elementary)

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## PROJECTS:

**Robotics**

- Designed and manufactured FIRST Robotics Competition robots using Solidworks and a variety of power tools (2016-2022)
- Led the CAD, Scouting, and Outreach teams during this time; Co-ran team social media
- Returned when in Austin to mentor the CAD and Outreach teams (since 2022)

**Scalable: A Rare-Scale Tuner and Visualizer**

- Web app deployed on AWS for tuning and visualization of sound, with less publicized scales
- Static version available at https://gummyhz.github.io/CSCI420_CloudComputing_Project3/ (learner account access ended)

**Plankton Distribution and Identification Website**

- Sampling and identification of plankton in Lake Matoaka at various locations with different influent flows
- Accessible at https://gummyhz.github.io/matoaka-plankton/

**Comparison and Recreation of Mycelial Growth Models**

- Reviewed literature on modelling of mycelial growth in hyphal fungi, with a focus on arbuscal mycorrhizae
- Recreated versions of differential equation models using my own implementation of direct numerical methods

**Decision Tree Models to Recognize Phishing Websites**

- Trained decision tree, random forest, and XGBoost models on a dataset of website characteristics to predict whether they were phishing websites or not
- Diagnostics and feature importances were extracted and compared for each model

**Linear Regression Modelling to Predict Bike Rentals**

- Used model selection and residual analysis to choose a best-performing linear regression model predicting bike rental counts based on environmental and temporal data

**Decision Tree and Neural Network Models to Recognize Disruptive Fusion Data**

- Using a dataset of fusion characteristic measurements, created models classifying fusion as disruptive or not
- Compared diagnostics and feature importances for decision tree, random forest, and XGBoost models
- Trained multi-layer perceptrons and bayesian neural networks and compared results